

Mathematical Foundations of Machine Learning

Week 5: Diagonalization

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1. Recap : Eigenvalues and eigenvectors
2. Diagonalization
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1 Quick Recap — Week 4

Definition (Linear Transformation). A map $T : V \rightarrow W$ between vector spaces over $\mathbb{R}$ is called a **linear transformation** if for all $u, v \in V$ and $\alpha \in \mathbb{R}$,

$$T(u + v) = T(u) + T(v), \quad T(\alpha v) = \alpha T(v).$$

Example (Differential Operator). Let $V = C^2([0, 1])$ and $W = C([0, 1])$, and define $T(f) = f'' + f'$. Then

$$T(\alpha f + \beta g) = \alpha(f'' + f') + \beta(g'' + g') = \alpha T(f) + \beta T(g),$$

so T is linear. In matrix form with respect to $\{e^x, e^{-x}\}$, T acts as

$$T(e^x) = 2e^x, \quad T(e^{-x}) = 0,$$

giving the matrix $A = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$ when we restrict T to the subspace $W = \text{Span}\{f_1(x) = e^x, f_2(x) = e^{-x}\}$.

Definition (Matrix Representation). For $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, define

$$T(e_j) = \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}, \quad A = (a_{ij}) \in \mathbb{R}^{m \times n}, \quad T(x) = Ax.$$

Then A is the **matrix of T** in the standard basis.

Example (Rotation–Scaling Map). Consider $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by

$$T(x, y) = (2x + y, x + 3y).$$

Then

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, \quad T(x) = Ax.$$

This transformation scales and rotates vectors simultaneously, with eigenvalues 1 and 4 corresponding to orthogonal eigenvectors.

Definition (Change of Basis). Let A be the standard matrix of $T : V \rightarrow V$. If P is the change-of-basis matrix from the standard basis to another basis B , then the matrix of T in B is

$$A_B = P^{-1}AP.$$

Example (Shear Map under Basis Change). Let $T(x, y) = (x + y, y)$ with $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. In the basis $B = \{(1, 1), (1, -1)\}$, we have

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad A_B = P^{-1}AP = \frac{1}{2} \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix}.$$

The matrix A_B shows that T acts as an asymmetric stretching along oblique axes.

Definition (Dual Space and Dual Map). The **dual space** of V , denoted V^* , is the set of all linear functionals $f : V \rightarrow \mathbb{R}$. Given $T : V \rightarrow W$, the **dual map** $T^* : W^* \rightarrow V^*$ is defined by

$$(T^*g)(v) = g(Tv).$$

Example (Integral Functional). Consider the linear operator

$$T : L^2([0, 1]) \rightarrow L^2([0, 1]), \quad (Tf)(x) = \int_0^x f(t) dt.$$

This is the classical Volterra integral operator.

Let $g \in (L^2([0, 1]))^*$ be the functional

$$g(h) = \int_0^1 x h(x) dx,$$

which corresponds (via the Riesz representation theorem) to the vector $x \in L^2([0, 1])$.

We now compute the dual map. By definition,

$$(T^*g)(f) = g(Tf) = \int_0^1 x (Tf)(x) dx = \int_0^1 x \left(\int_0^x f(t) dt \right) dx.$$

To simplify this expression, we use Fubini's theorem to change the order of integration. The region of integration is

$$\{(t, x) \in [0, 1]^2 : 0 \leq t \leq x \leq 1\},$$

so we may rewrite the integral as

$$\int_0^1 \int_t^1 x dx f(t) dt.$$

The inner integral is elementary:

$$\int_t^1 x dx = \left[\frac{x^2}{2} \right]_{x=t}^{x=1} = \frac{1}{2}(1 - t^2).$$

Therefore,

$$(T^*g)(f) = \int_0^1 f(t) \frac{1}{2}(1 - t^2) dt = \left\langle f, \frac{1}{2}(1 - t^2) \right\rangle_{L^2}.$$

Thus the adjoint T^*g corresponds, under the Riesz representation, to the vector

$$h(t) = \frac{1}{2}(1 - t^2) \in L^2([0, 1]).$$

Theorem 1.1 (Riesz Representation). *Let V be a finite-dimensional inner product space. For every $f \in V^*$, there exists a unique $y \in V$ such that*

$$f(x) = \langle x, y \rangle \quad \text{for all } x \in V.$$

Example (Riesz Representation in $\mathcal{P}_2$). Let $V = \mathcal{P}_2$ be the space of real polynomials of degree at most 2, equipped with the inner product

$$\langle p, q \rangle = \int_0^1 p(x) q(x) dx.$$

Consider the functional $f : V \rightarrow \mathbb{R}$ defined by

$$f(p) = p'(1).$$

By the Riesz representation theorem, there exists a unique $r \in \mathcal{P}_2$ such that for all $p \in V$,

$$p'(1) = \langle p, r \rangle = \int_0^1 p(x) r(x) dx.$$

We now determine r explicitly.

Step 1: Reduce $p'(1)$ to an integral formula. Since p is a polynomial, integrate by parts with $u = p$ and $dv = r(x) dx$:

$$\int_0^1 p'(x) r(x) dx = p(1)r(1) - p(0)r(0) - \int_0^1 p(x) r'(x) dx.$$

We want $\int_0^1 p(x) r(x) dx$ to equal $p'(1)$ for all p . Therefore we rewrite $p'(1)$ in a convenient integral form:

$$p'(1) = \int_0^1 p'(x) dx.$$

Step 2: Match coefficients for all p . From the integration by parts identity,

$$\int_0^1 p(x) r(x) dx = p(1)r(1) - p(0)r(0) - \int_0^1 p(x) r'(x) dx.$$

To represent $p'(1)$, we want

$$p'(1) = \int_0^1 p(x)r(x) dx = - \int_0^1 p(x)r'(x) dx \quad (\text{i.e. boundary terms vanish}).$$

Thus we impose the boundary conditions

$$r(1) = 0, \quad r(0) = 0.$$

Under these conditions,

$$p'(1) = - \int_0^1 p(x)r'(x) dx \quad \forall p.$$

To make this hold for all p , we must have

$$-r'(x) = \delta_{x=1}$$

in the sense of linear functionals on $\mathcal{P}_2$. Since $\mathcal{P}_2$ is finite-dimensional, this means that the map

$$p \mapsto p'(1)$$

must equal

$$p \mapsto - \int_0^1 p(x)r'(x) dx.$$

For **all quadratic polynomials**, we can identify the coefficient match by choosing the basis

$$p(x) = 1, \quad x, \quad x^2.$$

Step 3: Solve for coefficients of r .

Let

$$r(x) = a + bx + cx^2.$$

Then

$$r'(x) = b + 2cx.$$

We want

$$p'(1) = \int_0^1 p(x)r(x) dx,$$

for each basis element.

(i) For $p(x) = 1$:

$$p'(1) = 0, \quad \int_0^1 r(x) dx = a + \frac{b}{2} + \frac{c}{3}.$$

So:

$$a + \frac{b}{2} + \frac{c}{3} = 0.$$

(ii) For $p(x) = x$:

$$p'(1) = 1, \quad \int_0^1 xr(x) dx = \frac{a}{2} + \frac{b}{3} + \frac{c}{4}.$$

So:

$$\frac{a}{2} + \frac{b}{3} + \frac{c}{4} = 1.$$

(iii) For $p(x) = x^2$:

$$p'(1) = 2, \quad \int_0^1 x^2 r(x) dx = \frac{a}{3} + \frac{b}{4} + \frac{c}{5}.$$

So:

$$\frac{a}{3} + \frac{b}{4} + \frac{c}{5} = 2.$$

Solving this linear system gives

$$a = -2, \quad b = 8, \quad c = -6,$$

hence

$$r(x) = -6x^2 + 8x - 2.$$

Definition (Dual Norm). For a normed space $(V, \|\cdot\|)$, the **dual norm** on V^* is

$$\|f\|_* = \sup_{\|x\| \leq 1} |f(x)|.$$

Example (Dual of ℓ_p Norms). For $\|\cdot\|_p$ on $\mathbb{R}^n$, the dual norm is $\|\cdot\|_q$ with $1/p + 1/q = 1$. For instance, if $\|x\|_3 = (\sum_i |x_i|^3)^{1/3}$, then $\|y\|_* = \|y\|_{3/2}$. This follows from Hölder's inequality:

$$|\langle x, y \rangle| \leq \|x\|_3 \|y\|_{3/2}.$$

Definition (Eigenvalue and Eigenvector). Let $T : V \rightarrow V$. A nonzero $v \in V$ is an **eigenvector** if $T(v) = \lambda v$ for some scalar $\lambda \in \mathbb{R}$, the **eigenvalue** of T .

Example (Coupled Oscillator System). For $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ with

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix},$$

arising from a three-mass spring chain, we compute

$$\det(A - \lambda I) = (2 - \lambda)^3 - 4(2 - \lambda) + 2 = 0,$$

giving eigenvalues $\lambda_1 = 2 - 2 \cos \frac{k\pi}{4}$, $k = 1, 2, 3$. Eigenvectors correspond to standing-wave modes of the system.

Definition (Determinant of a Linear Transformation). For $T : V \rightarrow V$, define $\det(T) = \det([T]_{\mathcal{B}})$ for any ordered basis $\mathcal{B}$. This value is basis-independent.

Example (Volume Distortion in $\mathbb{R}^3$). Let $T(x, y, z) = (x + y, y + z, z + x)$. Then $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ and $\det(A) = 2$. Thus T doubles all oriented volumes, preserving orientation.

Definition (Characteristic Polynomial). For $T : V \rightarrow V$, the **characteristic polynomial** is

$$p_T(\lambda) = \det([T]_{\mathcal{B}} - \lambda I),$$

whose roots are the eigenvalues of T .

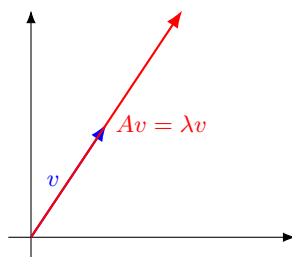


Figure 1: An eigenvector v retains its direction under A , only scaled by λ .

2 Eigenvalues, Eigenvectors and Diagonalization

2.1 Eigenvalues of Special Transformations

Theorem 2.1. If $T : V \rightarrow V$ is represented in some basis by an upper (or lower) triangular matrix, then its eigenvalues are precisely the diagonal entries of that matrix. Equivalently: the eigenvalues of T are the diagonal entries of $[T]_{\mathcal{B}}$ for any basis in which T is triangular.

Example. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by

$$T(x, y, z) = (x + 2y - 2z, 3y + 5z, -z).$$

Relative to the standard basis, the matrix is $\begin{bmatrix} 1 & 2 & -2 \\ 0 & 3 & 5 \\ 0 & 0 & -1 \end{bmatrix}$, which is upper triangular. Thus the eigenvalues are $1, 3, -1$.

2.2 Diagonalization as a Coordinate Simplification

Definition. A linear transformation $T : V \rightarrow V$ is said to be **diagonalizable** if there exists a basis $\mathcal{B}$ of V consisting of eigenvectors of T . In this basis, the representation $[T]_{\mathcal{B}}$ is diagonal, with the eigenvalues of T along the diagonal.

Example. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by $T(x, y) = (x + y, 2y)$. Its eigenvalues are $\lambda = 1, 2$ with eigenvectors $(1, 0)$ and $(1, 1)$. Thus, with respect to the basis $\{(1, 0), (1, 1)\}$, the matrix representation is diagonal:

$$[T]_{\mathcal{B}} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}.$$

Example (Non-Diagonalizable Operator). Let

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Then $p_A(\lambda) = (1 - \lambda)^3$, but $\dim \ker(A - I) = 1$, so A is not diagonalizable. Its Jordan canonical form reveals a chain of generalized eigenvectors, showing that T acts as a translation along a one-dimensional invariant subspace.

Theorem 2.2 (Diagonalizability criterion). *A linear transformation $T : V \rightarrow V$ is diagonalizable if and only if V has a basis consisting of eigenvectors of T .*

In summary, eigenvalues and eigenvectors are intrinsic to linear transformations. They do not depend on the choice of basis, although their computation is facilitated by choosing one. In an eigenbasis, the transformation is represented by a diagonal matrix, which reveals its action as simple scaling along invariant directions.

2.3 Diagonalization

An important consequence of eigenvalues and eigenvectors is the possibility of simplifying a matrix via diagonalization.

Definition. A matrix $A \in \mathbb{R}^{n \times n}$ is *diagonalizable* if there exists an invertible matrix P such that

$$P^{-1}AP = D,$$

where D is diagonal. In this case, the diagonal entries of D are the eigenvalues of A , and the columns of P are the corresponding eigenvectors.

Example. Let $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$. Take $P = I$, then $P^{-1}AP = A$ itself is diagonal.

For $A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$, the eigenvectors are $(1, 0)^\top$ and $(1, 1)^\top$. Choosing $P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, one computes $P^{-1}AP = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$. Thus A is diagonalizable.

Theorem 2.3 (Diagonalizability criterion). *If an $n \times n$ matrix has n linearly independent eigenvectors, then it is diagonalizable.*

Eigenvalues and eigenvectors thus provide a fundamental way of “decoding” the structure of a linear transformation. They highlight invariant directions, and their algebraic and geometric multiplicities determine whether a matrix can be simplified into a diagonal form.

3 Diagonalization

Definition. A matrix A is **diagonalizable** if there exists an invertible P such that

$$P^{-1}AP = D$$

with D diagonal.

Theorem 3.1. *If A has n linearly independent eigenvectors, then A is diagonalizable.*

Example. Consider the 3×3 matrix A :

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

The eigenvalues are the roots of the characteristic equation, $\det(A - \lambda I) = 0$.

$$\det(A - \lambda I) = \det \begin{pmatrix} 1 - \lambda & 1 & 0 \\ 0 & 2 - \lambda & 0 \\ 0 & 0 & 3 - \lambda \end{pmatrix} = 0$$

Since this is a triangular matrix (or we can use cofactor expansion along the first column), the determinant is the product of the diagonal entries:

$$(1 - \lambda)(2 - \lambda)(3 - \lambda) = 0$$

The eigenvalues are:

$$\lambda_1 = 1, \quad \lambda_2 = 2, \quad \lambda_3 = 3$$

Since all three eigenvalues are distinct, A has three eigenvectors, which must be linearly independent, and hence form a basis for $\mathbb{R}^3$. Therefore A is guaranteed to be diagonalizable.

To find the eigenvectors corresponding to each eigenvalue λ_i , we solve the homogeneous system $(A - \lambda_i I)v = 0$ for the eigenvector v .

Case $\lambda_1 = 1$. Solve $(A - 1I)v = 0$:

$$\begin{pmatrix} 1-1 & 1 & 0 \\ 0 & 2-1 & 0 \\ 0 & 0 & 3-1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

This system of equations simplifies to:

$$\begin{aligned} 0x_1 + x_2 + 0x_3 &= 0 &\implies x_2 &= 0 \\ 0x_1 + x_2 + 0x_3 &= 0 &\implies x_2 &= 0 \\ 0x_1 + 0x_2 + 2x_3 &= 0 &\implies x_3 &= 0 \end{aligned}$$

Let x_1 be the free variable, $x_1 = t$. The eigenvector is:

$$v_1 = t \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

We choose a simple eigenvector by setting $t = 1$:

$$v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

Case $\lambda_2 = 2$. Solve $(A - 2I)v = 0$:

$$\begin{pmatrix} 1-2 & 1 & 0 \\ 0 & 2-2 & 0 \\ 0 & 0 & 3-2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

The system of equations is:

$$\begin{aligned} -x_1 + x_2 &= 0 &\implies x_1 &= x_2 \\ x_3 &= 0 \end{aligned}$$

Let x_2 be the free variable, $x_2 = t$. Then $x_1 = t$ and $x_3 = 0$. The eigenvector is:

$$v_2 = t \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

We choose a simple eigenvector by setting $t = 1$:

$$v_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

Case $\lambda_3 = 3$. Solve $(A - 3I)v = 0$:

$$\begin{pmatrix} 1-3 & 1 & 0 \\ 0 & 2-3 & 0 \\ 0 & 0 & 3-3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -2 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

The system of equations is:

$$\begin{aligned} -2x_1 + x_2 &= 0 &\implies x_2 &= 2x_1 \\ -x_2 &= 0 &\implies x_2 &= 0 \end{aligned}$$

Substituting $x_2 = 0$ into the first equation gives $-2x_1 + 0 = 0$, so $x_1 = 0$. Let x_3 be the free variable, $x_3 = t$. The eigenvector is:

$$v_3 = t \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

We choose a simple eigenvector by setting $t = 1$:

$$v_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Then we write A with respect to the new ordered basis formed by the eigenvalues. The diagonal matrix D will have the diagonal entries as the eigenvalues :

$$D = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

P is formed by using the corresponding eigenvectors as columns (in the same order as their eigenvalues in D):

$$P = \begin{pmatrix} v_1 & v_2 & v_3 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

One may now easily check

$$A = PDP^{-1}$$

after computing P^{-1} - which is a good exercise to review your linear algebra notes.

Example. Consider the 3×3 matrix B :

$$B = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

We will show that this matrix is **not diagonalizable**. We follow the same steps. Namely, we first find the eigenvalues. The characteristic equation is $\det(B - \lambda I) = 0$:

$$\det(B - \lambda I) = \det \begin{pmatrix} 2 - \lambda & 1 & 0 \\ 0 & 2 - \lambda & 0 \\ 0 & 0 & 3 - \lambda \end{pmatrix} = 0$$

Since this is a triangular matrix, the determinant is the product of the diagonal entries:

$$(2 - \lambda)(2 - \lambda)(3 - \lambda) = 0$$

The eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = 3$. However, notice that the multiplicity of the eigenvalue 2 is 2.

- For $\lambda_1 = 2$, the factor $(2 - \lambda)$ appears twice.

$$\text{AM}(\lambda_1 = 2) = 2$$

- For $\lambda_2 = 3$, the factor $(3 - \lambda)$ appears once.

$$\text{AM}(\lambda_2 = 3) = 1$$

Now, let us find the Eigenvectors (v).

Case $\lambda_2 = 3$. Solve $(B - 3I)v = 0$:

$$\begin{pmatrix} 2 - 3 & 1 & 0 \\ 0 & 2 - 3 & 0 \\ 0 & 0 & 3 - 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

The system of equations is:

$$\begin{aligned} -x_1 + x_2 &= 0 &\implies x_1 &= x_2 \\ -x_2 &= 0 &\implies x_2 &= 0 \end{aligned}$$

Therefore, $x_1 = 0$ and $x_2 = 0$. x_3 is the free variable, $x_3 = t$.

$$v_2 = t \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

We get one linearly independent eigenvector:

$$v_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Case $\lambda_1 = 2$. Solve $(B - 2I)v = 0$:

$$\begin{pmatrix} 2-2 & 1 & 0 \\ 0 & 2-2 & 0 \\ 0 & 0 & 3-2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

The system of equations is:

$$\begin{aligned} 0x_1 + x_2 + 0x_3 &= 0 &\implies x_2 &= 0 \\ 0x_1 + 0x_2 + x_3 &= 0 &\implies x_3 &= 0 \end{aligned}$$

x_1 is the only free variable, $x_1 = t$.

$$v_1 = t \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

We get only one linearly independent eigenvector:

$$v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

Therefore, the eigenvectors do not form a basis of $\mathbb{R}^3$.

Definition (Algebraic Multiplicity). The **algebraic multiplicity** (AM) of an eigenvalue λ is its multiplicity as a root of the characteristic polynomial.

Example. We have seen that all the eigenvalues of

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

has multiplicity 1. However, for the matrix

$$B = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

the eigenvalue 2 has algebraic multiplicity 2.

Definition (Geometric multiplicity). The **geometric multiplicity** (GM) of an eigenvalue λ is the dimension of its eigenspace, $\dim(\text{Null}(B - \lambda I))$, which is the number of linearly independent eigenvectors associated with λ .

Example. Consider the matrix B above.

- For $\lambda_1 = 2$, we found only one linearly independent eigenvector.

$$\text{GM}(\lambda_1 = 2) = 1$$

- For $\lambda_2 = 3$, we found one linearly independent eigenvector.

$$\text{GM}(\lambda_2 = 3) = 1$$

The relationship between these two multiplicities determines diagonalizability:

1. **The Diagonalizability Condition:** A matrix is diagonalizable if and only if:

- The characteristic polynomial splits into linear factors (all eigenvalues are in the field, e.g., $\mathbb{C}$).
- For every eigenvalue λ , the Geometric Multiplicity equals the Algebraic Multiplicity ($\text{GM}(\lambda) = \text{AM}(\lambda)$).

2. **GM $\leq$ AM:** For any eigenvalue λ , the number of eigenvectors you can find is always limited by its multiplicity in the polynomial:

$$1 \leq \text{Geometric Multiplicity} \leq \text{Algebraic Multiplicity}$$

Example. In the matrix B above :

$$\text{AM}(\lambda = 2) = 2 \quad \text{but} \quad \text{GM}(\lambda = 2) = 1$$

The geometric multiplicity of 1 tells us that for the eigenvalue $\lambda = 2$, we are "missing" $2 - 1 = 1$ linearly independent eigenvector needed to form the full basis. The algebraic multiplicity of 2 is what we **hoped** to get, but the underlying structure of B (the presence of the 1 in the B_{12} position) forces the eigenspace to be only 1-dimensional.

The concept of geometric multiplicity is motivated by the need to count the actual dimensions of the eigenspaces, which is the true measure of how many basis vectors (eigenvectors) we have available to form the diagonalizing matrix P . When $\text{GM} < \text{AM}$, the matrix is not diagonalizable over that field.

4 Spectral Theorem

The Spectral Theorem focuses on a special and highly desirable type of diagonalization: one where the matrix P is an **orthogonal matrix**.

Definition (Orthogonal Matrix). A square matrix P is an **orthogonal matrix** if its columns (and rows) are orthonormal vectors, meaning:

$$P^T P = I \quad \text{or equivalently} \quad P^{-1} = P^T$$

Here are some properties of orthogonal matrices.

1. **Inverse is Transpose:** The inverse of an orthogonal matrix is simply its transpose: $P^{-1} = P^T$.
2. **Orthonormal Columns/Rows:** The columns of P form an **orthonormal basis** for $\mathbb{R}^n$. Similarly, the rows of P form an orthonormal basis for $\mathbb{R}^n$.
3. **Determinant Value:** The determinant of an orthogonal matrix must be 1 or -1 :

$$\det(P) = \pm 1$$

4. **Preserves Length (Norm):** For any vector $\mathbf{x}$, the length (Euclidean norm) of $\mathbf{x}$ is preserved under the transformation P :

$$\|P\mathbf{x}\| = \|\mathbf{x}\|$$

5. **Preserves Dot Product:** The dot product (inner product) between two vectors $\mathbf{x}$ and $\mathbf{y}$ is preserved:

$$(P\mathbf{x}) \cdot (P\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$$

6. **Eigenvalues:** If P has real eigenvalues, they must be 1 or -1 . In general, all eigenvalues of a real orthogonal matrix have an absolute value of 1 (i.e., they lie on the unit circle in the complex plane).

If a matrix A can be diagonalized by an orthogonal matrix P , it means A is particularly well-behaved:

$$A = PDP^{-1} = PDP^T$$

This is called **orthogonal diagonalization**.

The Spectral Theorem tells us exactly which matrices can be orthogonally diagonalized. The answer is simple and elegant: the matrix must be **symmetric**.

Theorem 4.1 (Spectral Theorem for Real Symmetric Matrices). *Let A be an $n \times n$ matrix with real entries. Then A is **orthogonally diagonalizable** if and only if A is a **symmetric matrix** ($A^T = A$). Furthermore, if A is a real symmetric matrix, then:*

1. *All eigenvalues of A are **real numbers**.*
2. *Eigenspaces corresponding to distinct eigenvalues are **orthogonal**.*
3. *A is always diagonalizable, meaning the "geometric multiplicity equals the algebraic multiplicity" for every eigenvalue.*

Example. The matrix $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ is real and symmetric, so the Spectral Theorem guarantees it is orthogonally diagonalizable.

Let's first find eigenvalues :

$$\det \begin{pmatrix} 1-\lambda & 2 \\ 2 & 1-\lambda \end{pmatrix} = (1-\lambda)^2 - 4 = 0$$

$$(1-\lambda)^2 = 4 \implies 1-\lambda = \pm 2$$

$$\lambda_1 = 1-2 = -1 \quad \text{and} \quad \lambda_2 = 1+2 = 3$$

Now, we find eigenvectors corresponding to the above eigenvalues :

Case $\lambda_1 = -1$. Solve $(A - (-1)I)\mathbf{v}_1 = 0$:

$$\begin{pmatrix} 1+1 & 2 \\ 2 & 1+1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

The equation is $2x_1 + 2x_2 = 0$, or $x_1 = -x_2$. Set $x_2 = 1$.

$$\mathbf{v}_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

Normalize $\mathbf{v}_1$:

$$\mathbf{u}_1 = \frac{1}{\|\mathbf{v}_1\|} \mathbf{v}_1 = \frac{1}{\sqrt{(-1)^2 + 1^2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

Case $\lambda_2 = 3$. Solve $(A - 3I)\mathbf{v}_2 = 0$:

$$\begin{pmatrix} 1-3 & 2 \\ 2 & 1-3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

The equation is $-2x_1 + 2x_2 = 0$, or $x_1 = x_2$. Set $x_2 = 1$.

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Normalize $\mathbf{v}_2$:

$$\mathbf{u}_2 = \frac{1}{\|\mathbf{v}_2\|} \mathbf{v}_2 = \frac{1}{\sqrt{1^2 + 1^2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Now, let's construct the matrices D and P .

$$D = \begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix}$$

$$P = (\mathbf{u}_1 \quad \mathbf{u}_2) = \begin{pmatrix} -1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}$$

The orthogonal diagonalization is $A = PDP^T$, which you are encouraged to verify.

Example. Consider the matrix C :

$$C = \begin{pmatrix} 1 & 3 & 3 \\ 0 & 5 & 6 \\ 0 & 0 & -2 \end{pmatrix}$$

First, eigenvalues; since C is an upper triangular matrix, the eigenvalues are the entries on the main diagonal:

$$\lambda_1 = 1, \quad \lambda_2 = 5, \quad \lambda_3 = -2$$

Since the eigenvalues are distinct, C is guaranteed to be diagonalizable.

Then, eigenvectors :

Case $\lambda_1 = 1$. Solve $(C - 1I)\mathbf{v}_1 = 0$:

$$\begin{pmatrix} 0 & 3 & 3 \\ 0 & 4 & 6 \\ 0 & 0 & -3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

The system is:

$$\begin{aligned} 3x_2 + 3x_3 &= 0 &\implies x_2 &= -x_3 \\ 4x_2 + 6x_3 &= 0 \\ -3x_3 &= 0 &\implies x_3 &= 0 \end{aligned}$$

If $x_3 = 0$, then $x_2 = -0 = 0$. x_1 is the free variable, $x_1 = t$.

$$\mathbf{v}_1 = t \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \implies \mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

Case $\lambda_2 = 5$. Solve $(C - 5I)\mathbf{v}_2 = 0$:

$$\begin{pmatrix} -4 & 3 & 3 \\ 0 & 0 & 6 \\ 0 & 0 & -7 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

The system is:

$$\begin{aligned} -4x_1 + 3x_2 + 3x_3 &= 0 \\ 6x_3 &= 0 &\implies x_3 &= 0 \\ -7x_3 &= 0 &\implies x_3 &= 0 \end{aligned}$$

Substitute $x_3 = 0$ into the first equation: $-4x_1 + 3x_2 = 0$, or $4x_1 = 3x_2$. Set $x_1 = 3$ and $x_2 = 4$.

$$\mathbf{v}_2 = \begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix}$$

Case $\lambda_3 = -2$. Solve $(C - (-2)I)\mathbf{v}_3 = 0$:

$$\begin{pmatrix} 3 & 3 & 3 \\ 0 & 7 & 6 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

The system is:

$$\begin{aligned} 3x_1 + 3x_2 + 3x_3 &= 0 &\implies x_1 + x_2 + x_3 &= 0 \\ 7x_2 + 6x_3 &= 0 \end{aligned}$$

From the second equation, $7x_2 = -6x_3$. Set $x_3 = 7$ and $x_2 = -6$. Substitute into the first equation: $x_1 + (-6) + 7 = 0 \implies x_1 + 1 = 0 \implies x_1 = -1$.

$$\mathbf{v}_3 = \begin{pmatrix} -1 \\ -6 \\ 7 \end{pmatrix}$$

Then, we obtain :

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$

$$P = (\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3) = \begin{pmatrix} 1 & 3 & -1 \\ 0 & 4 & -6 \\ 0 & 0 & 7 \end{pmatrix}$$

The diagonalization is $C = PDP^{-1}$.